ZCAS University

CCS1501 ENGINEERING MATHEMATCIS

FINAL EXAMINATION

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. **Section A**: this question is **compulsory** and must be attempted and it carries **40 Marks**.
2. **Sections B**: Answer **ANY** three (3) questions from this section.
3. This question paper carries a total of **100 marks**.
4. Calculators are **NOT ALLOWED**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

SECTION A: Question 1 is compulsory and must be attempted

Question one

(a) Given the set $P = \{x: 0 \leq x \leq 3, x \epsilon Z\}$ and $Q = \{x: 2 < x \leq 4,\ x \epsilon Z\}$, where P and Q are subsets of

$M = \{x: 0 \leq x \leq 10, x \epsilon Z\}$. List

(i) $P \cap Q$ **[2 Marks]**

(ii) $Q \cup P$ **[2 Marks]**

(iii) $P \times Q$ **[2 Marks]**

(b) (i) Express as a single logarithm $2\log_4(2x+3) - \log_4 x - \log_4(2x-1)$ **[1 Mark]**

(ii) Hence, solve the equation $2\log_4(2x+3) = 1 + \log_4 x + \log_4(2x-1)$ **[3 Marks]**

(c) Solve the equation $4(3^{2x+1}) + 17(3^x) - 7 = 0$ **[4 Marks]**

(d) Write down the first three terms in the Binomial expansion of $\sqrt{1+x}$ **[3 Marks]**

(e) Decompose $\frac{x}{(x-3)^2}$ into partial fractions. **[4 Marks]**

(f) Given that $f(x) = ax^3 + (a+b)x^2 + (a+2b)x + 1$ is exactly divisible by $(x+1)$.

(i) Express b in terms a. **[3 Marks]**

(ii) Use synthetic division to find the **quotient** when $f(x)$ is divided by $(x+1)$.

[3 Marks]

(g) (i) Find the vector equation, parametric equations for the line whose Cartesian equation is

$\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$ **[3 Marks]**

(ii) The vector a line is $r = i - 2j + 2k + \lambda(5\mathrm{i} + 2\mathrm{j} - \mathrm{k})$ express the equation of this line in parametric form and hence, find the coordinates of points where the line crosses the $xy - plane$, find the direction cosines of the direction vector. **[5 Marks]**

(iii) A line has a Cartesian equations $\frac{x-1}{3} = \frac{y+2}{4} = \frac{z-4}{5}$, find a vector equation for a parallel line passing through the point with position vector $5i = 2j - 4k$ and find the coordinates of the point on this line where $y = 0$. **[5 Marks]**

[TOTAL: 40 Marks]

SECTION B: Attempt ANY THREE questions in this section

Question 2

(a) Solve the equation $|2x - 5| = 7x$ **[2 Marks]**

(b) Write the fourth term in the expansion of $\left(px + \frac{q}{x}\right)^n$ in its simplest form. Find the value of n if the fourth term is independent of x. With this value of n, calculate the values of p and q given that the fourth term is 160 and $p - q = 1$, where $p, q \in Z^+$ **[6 Marks]**

(c) Given that $f(x) = 2x^3 - x^2 - 5x - 2$

(i) Show that $f(x)$ is divisible by $x - 2$. **[2 Marks]**

(ii) Hence, solve the inequality $f(x) \leq 0$ **[5 Marks]**

(d) If $f(x) = 6x - 2$ and $g(x) = x^2 - 2x + 5$. Find $g[f(-1)]$ **[5 Marks]**

[Total: 20 Marks]

Question 3

(a) Given the equation $ax^2 + bx + c = 0$, show that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ **[5 Marks]**

(b) The equation $3x^2 - 7x - 6 = 0$ has roots α and β. Find the quadratic equation with integer coefficients whose roots are $\alpha - 5$ and $\beta - 5$ **[6 Marks]**

(c) The expression $\frac{3x+7}{x^3+x^2-2x}$ can be resolved into sums of partial fractions by firstly finding the factors of its denominator. Determine these factors. **[4 Marks]**

(d) (i) Define a one-to-one function **[2 Marks]**

(ii) Show that $f(x) = 9x - 2$ is a one to one function **[3 Marks]**

[Total: 20 Marks]

Question 4

(a) Find the sixth (6th) term in the expansion $(2x - 3y)^9$ **[5 Marks]**

(b) Expand the following by the binomial formula $(2x - y)^4$ **[5 Marks]**

(c) Use the binomial formula to compute $(1.02)^4$ to two decimal places. **[5 Marks]**

(d) Find the middle term of the expansion $\left(3x^2 + \frac{1}{2x}\right)^{10}$ **[5 Marks]**

[Total: 20 Marks]

Question 5

(a) Factorize $f(x) = 2x^3 - 3x^2 - 8x + 12$ **[5 marks]**

(b) Given that $Z = \begin{pmatrix} 9 & 7 & 3 \\ 5 & -1 & 4 \\ 6 & 8 & 2 \end{pmatrix}$

(i) Calculate Z^2 **[3 Marks]**

(ii) Find Z^{-1}, and hence, **[7 Marks]**

(iii) Solve the following systems of equations

$$9x + 7y + 3z = 6$$
$$5x - y + 4z = 1$$
$$6x + 8y + 2z = 4$$ **[5 Marks]**

[Total: 20 Marks]

END OF EXAMINATION!!